

The altitude extension of the mid-latitude trough and its correlation with plasmopause position

Endawoke Yizengaw and Mark B. Moldwin

Institute of Geophysics and Planetary Physics, University of California,
Los Angeles, California, USA

Abstract

[1] Tomographic imaging provides a powerful technique for obtaining images of the spatial distribution of ionospheric electron density in an altitude versus geographic latitude grid at a specified longitudinal sector. The method, which involves monitoring the total electron content (TEC) of the ionosphere using signals transmitted from the global positioning system (GPS), has the ability to reveal the detailed structure of the ionosphere including the mid-latitude trough. The tomographic reconstruction of ionospheric electron density structures are performed at different specific longitudinal sectors both in the northern and southern hemisphere. At the same time the plasmopause position is estimated from IMAGE EUV images and mapped onto tomographic images. The simultaneous location of the altitude extension of the mid-latitude trough and the plasmopause position are essentially co-located.

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